MAgent Theory 2.0

A Physicist's Companion

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October 2026

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This companion develops interacting statistical agents and inherited organization through worked examples. It assumes probability, linear algebra, and differential equations. The examples distinguish nonsmooth fusion from conditional reductions under the original smooth dynamics; the surrounding theory permits general probability laws.

Chapter 1

What an Agent Carries

A classical statistical description distinguishes a phase-space point from a distribution over phase space. The same distinction begins this theory: $k \in \mathsf{K}$ is a represented possibility, while q is a probability law over such possibilities. The agent carries the law and mechanisms for changing it. Its contents need not be positions and momenta, and the statistical law need not describe an ensemble of physically separate agents.

1.1 Beliefs, models, and predictions

There are two sectors. The belief law q_i describes state contents k. The model law s_i describes model presentations $m \in \mathsf{M}$. Their most general joint description is

$$\zeta_i(dk, dm) = q_i(dk \mid m) s_i(dm). \tag{1.1}$$

The notation $q_i(dk \mid m)$ means a conditional law, and $q_i(dk)$ without the condition denotes its marginal. Choosing $\zeta_i = q_i \otimes s_i$ removes dependence between the sectors in that description. It still permits coupled equations for q_i and s_i, just as coupled equations for two coordinates do not require a statistical ensemble over those coordinates.

A model presentation becomes predictive through a normalized evaluator $G_i(m, dk)$. For each m, this assigns a probability law to the state contents. Averaging the evaluator over the model law gives

$$(s_i G_i)(A) = \int G_i(m, A) s_i(dm). \tag{1.2}$$

For example, let k, m be binary, with $s_i(m = 1) = 0.6$, $G_i(1, \{1\}) = 0.9$, and $G_i(0, \{1\}) = 0.2$. The model mixture assigns probability 0.62 to $k = 1$. A present belief $q_i(k = 1) = 0.8$ is a different object. An interaction rule can bring these descriptions into agreement, revise the model law, or change both. Those dynamics must be specified.

An evaluator is a Markov kernel: a function whose output is a probability measure. Ordinary functions are included as the case in which the output measure is concentrated at one value. This language accommodates discrete, continuous, singular, and multimodal laws without selecting a Gaussian family. When differentiating laws later, we will restrict to a regular statistical family or positive finite tables.

1.2 A record has an address inside the agent

Let $x_i \in \mathsf{S}_i$ denote the complete internal state, including its laws, interfaces, and any memory. A record readout is an ordinary measurable function

$$\rho_i : \mathsf{S}_i \longrightarrow \mathsf{O}_i. \tag{1.3}$$

The output alphabet O_i specifies the distinctions the agent can read. A binary register gives $\mathsf{O}_i = \{0, 1\}$; a real-valued comparison can use $\mathbb{R}^r$. The detector analogy is precise at this level: an internal state contains a readable distinction. It does not require an outside observer to inspect it.

Write $x \in \mathsf{S}$ for the complete population configuration. A deterministic interaction $F_{ij} : \mathsf{S} \to \mathsf{S}$ produces the record

$$o_{ij} = \rho_i\big(\mathrm{pr}_i\, F_{ij}(x)\big), \tag{1.4}$$

where pr_i selects the receiver's state. For a stochastic transition $\Gamma_{ij}(x, dx')$, the probability of a record in A is

$$\mathcal{R}_{ij}(x, A) = \Gamma_{ij}\big(x, \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}\big). \tag{1.5}$$

This is a pushforward: collect the probability of every next state that displays an outcome in A. The transition specifies what happens; the readout specifies what is accessible. A persistent record requires that some later readout still recover the distinction. A transient comparison and a memory trace therefore have different dynamical requirements.

A likelihood on represented contents has another role. For instance, the expression $o = A_i k_i - B_{ij} k_j + \epsilon$ gives a comparison value once the contents and noise are given. Their joint law predicts possible comparisons. A registration rule must still place the outcome in an internal state and specify the response to it.

1.3 Which probability space is being used?

The configuration x contains the agents' laws. A tuple of represented contents $(k_i, m_i)_i$ belongs to a different space. A probability law over possible configurations is a third construction, useful when a preparation or stochastic dynamics has been declared. Deterministic motion of x requires neither sampling the contents nor drawing an ensemble of configurations at each step.

This distinction also clarifies correlations. Two independent fair bits have probability $1/4$ for each pair. Condition on a shared record that the bits agree. Their marginals remain fair, but the conditional joint law has probability $1/2$ on each of $(0, 0)$ and $(1, 1)$. A calculation of the next joint comparison needs this dependence; the marginals do not contain it. A coupled product-family evolution remains a possible model, but it will not represent that exact conditional joint law. Whether a correlation is needed is decided by the response being predicted.

The laws can be attached to local domains of a context base $\mathcal{C}$. This is the section picture: a context selects the law applicable there, and a frame specifies its coordinates. Most interactions below occur at a common context. Moving between contexts introduces the connection problem in chapter 8.

Chapter 2

Frames, Messages, and Interactions

An agent's probability law and the coordinates used to describe that law are different objects. Communication introduces a third object: a rule for translating, filtering, or modifying what another agent supplies. Keeping these distinctions explicit prevents a coordinate change from being mistaken for an interaction, or a selective message from being mistaken for an invertible transport.

2.1 Reading a link from source to receiver

There are two sectors. Write $\nu_i^q = q_i$ for the belief marginal and $\nu_i^s = s_i$ for the model marginal. In a finite-dimensional realization, a sector $a \in \{q, s\}$ has a reference vector space V_a, an agent space E_i^a, and an invertible presentation

$$U_i^a : V_a \longrightarrow E_i^a, \qquad \bar{\nu}_i^a = (U_i^a)_{\#}^{-1} \nu_i^a. \tag{2.1}$$

The pushforward notation expresses the represented possibilities in reference coordinates. The two sectors may use representations of the same frame group while having different dimensions, laws, and interaction rules.

A law-dependent reference generator and its displayed link are

$$L_{ij}^a = \mathcal{L}_{ij}^a(\bar{\nu}_i^a, \bar{\nu}_j^a), \qquad \Theta_{ij}^a = U_i^a e^{L_{ij}^a} (U_j^a)^{-1}. \tag{2.2}$$

The index order is receiver first, source second. Reading from right to left, the source is expressed in reference coordinates, the generator acts there, and the result is expressed at the receiver. The function $\mathcal{L}_{ij}^a$ is a constitutive choice. The exponential parameterization does not derive it or equate the belief and model generators.

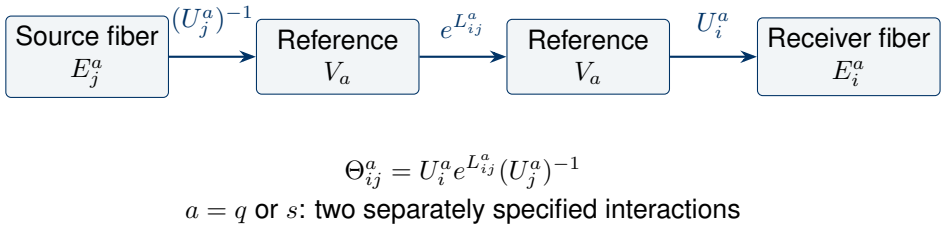


Figure 2.1: The source presentation is undone by $(U_j^a)^{-1}$, the interaction $e^{L_{ij}^a}$ acts within the reference space, and U_i^a presents the result at the receiver. Endpoint frames and the intervening interaction have separate roles.

2.2 Changing coordinates

For a passive change $y_i' = g_i^a y_i$, both the law and its presentation change: $\nu_i^{a\prime} = (g_i^a)_{\#}\nu_i^a$ and $U_i^{a\prime} = g_i^a U_i^a$. The reference law stays fixed, giving

$$\Theta_{ij}^{a\prime} = g_i^a \Theta_{ij}^a (g_j^a)^{-1}. \tag{2.3}$$

This endpoint covariance is familiar from gauge link variables [9]. Metrics and additional model data must also be re-expressed when forming scalar predictions. Covariance alone supplies no physical symmetry or evolution law.

Changing the common reference presentation is a separate operation. Under $U_i^a \mapsto U_i^a h_a$, the reference laws become $(h_a^{-1})_{\#}\bar{\nu}_i^a$. Reference consistency requires the constitutive rule, including any interface data, to give $L_{ij}^{a\prime} = h_a^{-1} L_{ij}^a h_a$. The identity $e^{h_a^{-1} L h_a} = h_a^{-1} e^L h_a$ then leaves the displayed link unchanged.

2.3 Selective interfaces and an explicit pair

Sending and receiving can be represented by

$$T_j^a : V_a \longrightarrow H_a, \qquad O_i^a : H_a \longrightarrow V_a, \qquad L_{ij}^a = O_i^a T_j^a. \tag{2.4}$$

The sender T_j^a extracts a message; the receiver O_i^a inserts its effect. A direct content map $O_i^a T_j^a$ can lose dimensions. Its exponential is always invertible and acts as identity on $\ker T_j^a$. Thus exponentiating a selective generator retains unread directions. Sending only the extracted message is a different operation. General normalized Markov kernels allow communication without any linear realization.

One explicit law-dependent choice uses finite second moments and positive-definite covariance. Let μ_i^a, M_i^a be the reference mean and covariance. On nonself edges choose a scalar λ_{ij}^a and set

$$L_{ij}^a = \lambda_{ij}^a \mu_i^a (\mu_j^a)^T (M_j^a)^{-1}. \tag{2.5}$$

The row $(\mu_j^a)^T (M_j^a)^{-1}$ sends a scalar message; the column $\lambda_{ij}^a \mu_i^a$ receives it. Under a reference change, the mean transforms by h_a^{-1} and covariance by congruence. Their product consequently transforms by conjugation. These assumptions concern moments, not Gaussian shape. Primitive self-comparisons are set to identity separately.

For a concrete non-Gaussian example, let Z have independent centered Laplace coordinates of variance one in two dimensions. Give two agents the laws of $e_1 + Z$ and $e_2 + Z$, with identity frames and unit coefficients. Then $M_1 = M_2 = I$, $L_{12} = E_{12}$, and $L_{21} = E_{21}$. Since each generator squares to zero,

$$\Theta_{12}\Theta_{21} = (I + E_{12})(I + E_{21}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{2.6}$$

Here E_{ab} has its only nonzero entry, one, in row a, column b. The directed communication loop is nontrivial. It establishes neither a connection over contexts nor a physical curvature.

Chapter 3

Coupled Relaxation and Source Selection

The dynamical variables are probability laws. Moving an agent therefore means changing its expectations, uncertainties, and dependencies, rather than moving a point through its sample space. The physical analogue developed here is overdamped motion on a probability manifold: an interaction potential supplies the driving covector, and the Fisher metric converts that covector into a velocity. Neither ingredient determines the other.

3.1 The potential and its source rows

At a fixed context, let q_i describe agent i's represented state k, and let s_i describe its model presentation m. For this construction their joint recognition law is the explicit product $q_i \otimes s_i$. Choose a model reference $r_i(dm)$ and a normalized evaluator $G_i(m, dk)$. Their local contribution is

$$\mathcal{E}_0 = \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| r_i G_i) = \sum_i \left[D_{\mathrm{KL}}(s_i \| r_i) + \int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot)) s_i(dm) \right]. \tag{3.1}$$

The equality separates the model marginal from the conditional state law. Thus a product recognition family still has coupled dynamics: changing s_i changes the evaluator against which q_i relaxes, and changing q_i changes the model costs.

The channels of chapter 2 send peer laws into comparable spaces. Write $\nu^q_{ij} = q_j \mathcal{K}^q_{ij}[\mathbf{q}]$ and $D^q_{ij} = D_{\mathrm{KL}}(q_i \| \nu^q_{ij})$, where $\mathbf{q}$ collects the belief laws. Define ν^s_{ij} and D^s_{ij} similarly for model laws. With normalized source rows β_i, γ_i, positive reference rows π^q_i, π^s_i, and $T_q, T_s > 0$, the potential is

$$\begin{aligned} \mathcal{E} = \mathcal{E}_0 &+ \sum_i \left[\sum_j \beta_{ij} D^q_{ij} + T_q D_{\mathrm{KL}}(\beta_i \| \pi^q_i) \right] \\ &+ \sum_i \left[\sum_j \gamma_{ij} D^s_{ij} + T_s D_{\mathrm{KL}}(\gamma_i \| \pi^s_i) \right]. \end{aligned} \tag{3.2}$$

Each sum uses its declared source set. The row penalty balances preference for compatible peers against departure from the reference routing.

Hold the laws fixed. For either sector, minimize $f(\eta) = \sum_j \eta_j D_j + T D_{\mathrm{KL}}(\eta\|\pi)$ subject to $\sum_j \eta_j = 1$. Stationarity gives $D_j + T[\log(\eta_j/\pi_j) + 1] = \lambda$, hence

$$\eta_j^* = \frac{\pi_j e^{-D_j/T}}{Z}, \qquad Z = \sum_j \pi_j e^{-D_j/T}, \qquad f(\eta) = -T \log Z + T D_{\mathrm{KL}}(\eta\|\eta^*). \tag{3.3}$$

The last identity proves that this stationary row is the unique minimum. Eliminating both sets of rows gives $\mathcal{E}_{\mathrm{red}} = \mathcal{E}_0 - T_q \sum_i \log Z_i^q - T_s \sum_i \log Z_i^s$. At fixed temperatures and reference rows,

$$d\mathcal{E}_{\mathrm{red}} = d\mathcal{E}_0 + \sum_{ij} \beta_{ij}^* dD_{ij}^q + \sum_{ij} \gamma_{ij}^* dD_{ij}^s. \tag{3.4}$$

This follows by differentiating each log partition sum. Optimized row derivatives disappear; derivatives inside their costs remain.

3.2 Every interaction role contributes to motion

For normalized densities a, b, variation of $\int a \log(a/b)$ gives

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b) da - \int (a/b) db, \qquad d\nu_{ij}^q = (dq_j)\mathcal{K}_{ij}^q + q_j D\mathcal{K}_{ij}^q(d\mathbf{q}). \tag{3.5}$$

Here da, db denote mass-preserving variations, and differentiation assumes a regular dominated family. An agent moves through its receiver role, its source role, and any dependence of the channels on its law. Directed routing does not remove incoming contributions to source motion. For an identity self-channel, the two laws remain identical and their divergence stays zero. Holding an old self snapshot fixed instead creates a different potential after the current law changes.

For positive finite probability tables, collect all dynamical tables as u^a, including the source rows when they are not eliminated. The Fisher metric and its unit-mobility descent are

$$\begin{aligned} g_u(v,v) &= \sum_{a,z} \frac{(v_z^a)^2}{u_z^a}, \\ \dot{u}_z^a &= -u_z^a(h_z^a - \mathbb{E}_{u^a} h^a), \qquad h_z^a = \frac{\partial \mathcal{E}}{\partial u_z^a}, \\ \dot{\mathcal{E}} &= -\sum_a \mathrm{Var}_{u^a}(h^a). \end{aligned} \tag{3.6}$$

Centering preserves each normalization. Substituting the velocity into $\dot{\mathcal{E}} = \sum_{a,z} h_z^a \dot{u}_z^a$ proves the dissipation identity. The same construction uses the Fisher tensor on a regular smooth family [1]. Both sectors have equal mobility; a slower model response must arise from the resulting dynamics.

A Bernoulli law makes the metric concrete. Its success probability p has metric $dp^2/[p(1-p)]$, so the mobility is $p(1-p)$. For $\mathcal{E}(p) = D_{\mathrm{KL}}(\mathrm{Ber}(p)\|\,\mathrm{Ber}(p_0))$, with $p, p_0 \in (0,1)$, write $l = \log[p/(1-p)]$ and $l_0 = \log[p_0/(1-p_0)]$. Then

$$\dot{p} = -p(1-p)(l - l_0), \qquad \dot{l} = -(l - l_0). \tag{3.7}$$

Log odds relax exponentially, although probability does not follow a linear restoring equation. Here t parametrizes the chosen relaxation, and the dimensionless T_q, T_s weight routing penalties. They have not been identified with physical time or bath temperature.

3.3 A record likelihood defines a different objective

A fixed normalized prior P_0 and record likelihood ℓ_o give posterior $\Pi_o = \ell_o P_0 / Z_o$ and the exact identity

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| P_0) - \mathbb{E}_Q \log \ell_o = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o. \tag{3.8}$$

Source inference in a product recognition family then uses expected negative log likelihood as its cost. Peer coupling instead uses divergence between laws, whose source template can move during relaxation. Both yield Gibbs rows because both contain the same categorical relative-entropy penalty; their costs and state derivatives differ. Identical independent Gaussian laws have zero peer divergence but nonzero expected squared sample difference. A normalized likelihood for that difference therefore describes an interaction distinct from direct agreement of the laws.

Chapter 4

What a Message Lets the Receiver Infer

Sending a few observables can change an agent's description of many variables. The extra structure comes from the receiver's conditional model. This is familiar in physics: measuring one coordinate of a correlated system can change predictions for another coordinate without measuring it directly. The distinction between transmitted information and inferred structure is explicit here.

4.1 Selective sending and conditional reconstruction

A sender uses a measurable map $T_j : E_j \to \mathsf{H}$ into a message space. If its content has law q_j, possible messages have law

$$\nu_j = (T_j)_{\#} q_j. \tag{4.1}$$

The pushforward symbol means $\nu_j(A) = q_j(T_j^{-1}A)$. Sampling a message and transmitting the whole law ν_j are different communication protocols. A linear receiving map $O_i : \mathsf{H} \to E_i$ embeds a message, but its composition with T_j cannot have higher rank than T_j.

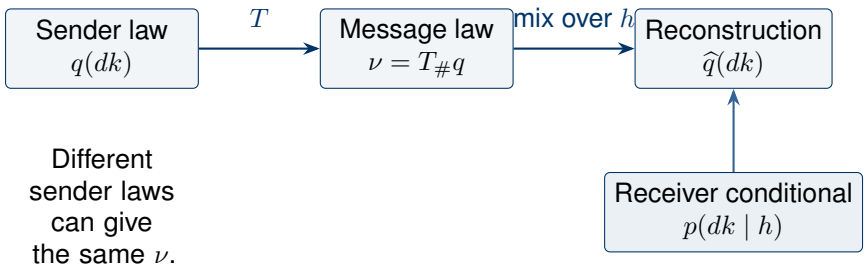


Figure 4.1: Selective communication and receiver structure. The sender fixes the transmitted marginal. The receiver supplies a conditional law for distinctions that the message does not specify. This distributional reconstruction presupposes that the message law is available; one received sample supplies a value instead.

To see exactly what is lost, work in one common presentation and write $T : E \to \mathsf{H}$. Let q be the sender's law and p the receiver's reference law, with finite $D_{\mathrm{KL}}(q\|p)$. The receiver can combine the transmitted marginal $\nu = T_{\#}q$ with its own conditional law:

$$\widehat{q}(dk) = \int p(dk \mid h)\nu(dh). \tag{4.2}$$

Disintegration on standard Borel spaces provides these conditional laws [4]. The reconstruction has exactly the right message marginal. Its remaining discrepancy is

$$D_{\mathrm{KL}}(q\|\widehat{q}) = \int D_{\mathrm{KL}}\big(q(\cdot \mid h)\|p(\cdot \mid h)\big)\nu(dh). \tag{4.3}$$

Both laws assign the same probability to h, so that factor cancels in their density ratio. The remaining logarithm compares the two conditional laws, which gives the equality by integration.

For a calculation on four points, let $k = (H, V)$ be two bits. The sender gives probability $1/2$ to each of 00 and 11, and transmits only H. A receiver whose reference makes both bits independent and fair reconstructs the uniform law on all four points. The message marginal is correct, while the relative entropy is $\log 2$. A receiver whose conditional model has $V = H$ reconstructs the sender's full law. The missing correlation was supplied by the receiver.

For a single received value h, Bayesian assimilation instead takes the form

$$q_i^+(dk) = \frac{\ell_i(h \mid k) q_i^-(dk)}{\int \ell_i(h \mid k') q_i^-(dk')}, \tag{4.4}$$

with a positive finite denominator. In the perfectly correlated two-bit example, a noiseless observation $H = 1$ also fixes $V = 1$. This is inference through an existing relation, not an inversion of an arbitrary lossy map.

4.2 Agreement needs a common prediction rule

Suppose a model law s and a normalized record evaluator $K(m, do)$ predict the record law sK. If two agents compare model laws in one frame, they must also compare their evaluators in that frame. Equal model laws can predict opposite bits when their evaluators differ.

For one common evaluator and finite $D_{\mathrm{KL}}(s\|t)$, an exact information balance is

$$D_{\mathrm{KL}}(s\|t) = D_{\mathrm{KL}}(sK\|tK) + \int D_{\mathrm{KL}}(s^o\|t^o)(sK)(do), \tag{4.5}$$

where s^o, t^o are the conditional model laws after record o. The joint laws $s(dm)K(m, do)$ and $t(dm)K(m, do)$ have divergence $D_{\mathrm{KL}}(s\|t)$ because their common kernel cancels. Factoring the same joint laws by their record marginal gives the displayed split. The last term is the distinction left unresolved by the record. A constant evaluator discards every distinction; a sufficient record leaves none.

For a model link Θ_{ij}^s, compatibility means $K_i(\Theta_{ij}^s m, A) = K_j(m, A)$ after identifying record alphabets. Only then does transporting the model law also transport its predictive meaning. If the link changes with the laws, this condition must continue to hold at the updated state.

4.3 Compression should name what matters

Rate–distortion theory makes the desired fidelity part of the communication problem [6]. A low-dimensional real vector alone is not a finite-bit channel. Noise, precision, or an encoding constraint must specify the communication cost. Wyner–Ziv coding additionally gives the receiver a realized variable correlated with the source [10]; possession of a prior by itself is not that realized side information.

For a predictive criterion, name the source X, target response Y, and accessible receiver information Z. An encoder generates U from X alone, so $U - X - (Y, Z)$ is a conditional-independence

chain. The optimal decoder is $P(Y \mid U, Z)$. Its expected relative-entropy distortion is

$$\mathcal{D} = I(Y; X \mid U, Z) = I(Y; X \mid Z) - I(Y; U \mid Z). \tag{4.6}$$

The equality follows by expanding $I(Y; X, U \mid Z)$ in both orders. It is the information about the chosen response that compression discards. An information-bottleneck criterion trades this quantity against a cost such as $I(X; U \mid Z)$ [7]; an operational bit rate requires additional coding assumptions.

If $X = (A, B)$ consists of independent fair bits and $U = A$, the distortion is zero for target $Y = A$ and $\log 2$ for target $Y = B$. For a prospective parent, the target can be a future boundary response. Retaining one such response does not ensure that every later interaction can be inherited. That stronger requirement concerns dynamics.

Chapter 5

When a Collective Can Act as an Agent

A collective can have a simple description without being able to evolve from that description alone. Knowing the total energy of two bodies, for example, does not usually determine their next interaction. A parent of law-valued agents faces the same question: does its retained state determine its future responses?

There are two operations to distinguish. A channel C compresses represented contents and gives a parent law $Q_x C$. A map R retains a complete dynamical state $R(x)$. Besides belief and model laws, that state can include the model evaluator, source information, boundary channels, and memory. A distribution over possible contents does not automatically contain these additional data.

5.1 Two routes to the same parent

Suppose the fine configuration advances by F. Exact inherited evolution requires a parent update F^c such that

$$R(F(x)) = F^c(R(x)). \tag{5.1}$$

We can first evolve the children and then form their parent, or first form the parent and evolve it. Both routes must give the same retained state. Repeating the identity gives agreement at every later step. Boundary response mechanisms must likewise depend only on the retained state.

The immediate test is whether configurations with the same parent have the same next parent. If $R(x) = R(x')$ but $R(F(x)) \neq R(F(x'))$, no single F^c can satisfy the equation. Conversely, in a finite state space, this test is sufficient: evolve any representative of each parent state. Agreement within each fiber of R makes the choice irrelevant. On general measurable spaces the resulting map must also be measurable.

For stochastic dynamics, replace the next state by its distribution. If T is the fine transition matrix, the probability

$$\sum_{y:R(y)=z'} T(x, y) \tag{5.2}$$

must depend only on $R(x)$. These sums then define normalized parent rows. Conditioning on any parent history merely averages identical rows, so the parent reproduces the whole retained history law. This is strong lumpability [5]. Agreement for one equilibrium preparation is a weaker statement.

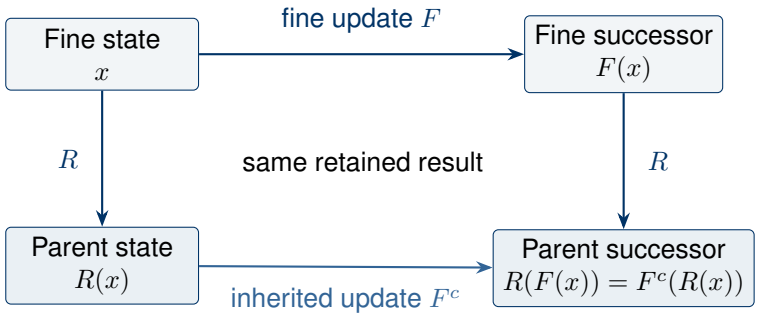


Figure 5.1: The two routes to the next parent configuration must agree. Vertical maps retain information; horizontal maps advance the dynamics. A compressed law alone need not make this square commute.

5.2 A missing role can prevent closure

Let two independent represented bits have laws $\mathrm{Ber}(p_1)$ and $\mathrm{Ber}(p_2)$. Their probabilities evolve according to

$$\begin{aligned} \dot{p}_1 &= -p_1 + \lambda(p_2 - p_1), \\ \dot{p}_2 &= -2p_2 + \lambda(p_1 - p_2), \qquad \lambda \geq 0. \end{aligned} \tag{5.3}$$

The square $[0,1]^2$ is preserved: the derivative points inward at every boundary. Retain only the law of the sum of the two bits. The configurations $(0.8, 0.2)$ and $(0.2, 0.8)$ both give probabilities $(0.16, 0.68, 0.16)$ for sums $0, 1, 2$.

Their subsequent changes differ. The coupling terms cancel when we add the two equations, giving

$$\frac{d}{dt}\mathbb{E}(K_1 + K_2) = -(p_1 + 2p_2). \tag{5.4}$$

The derivatives are -1.2 and -1.8. Thus the same retained law has two different possible derivatives. Compression erased which probability belonged to the faster-decaying role. The fine recognition law remains a product throughout. What must be retained is the information this evolution uses; cross-agent covariance is not automatically required.

5.3 An exact supplied hierarchy

A finite construction shows that two successive reductions can work. Take $x = (a, b, c, r) \in \{0,1\}^4$, with internal register r and readout $\rho(x) = r$. Let the agent laws be

$$q_a = \mathrm{Ber}\left(\frac{1+2a}{4}\right), \qquad s_a = \mathrm{Ber}\left(\frac{2+a}{4}\right). \tag{5.5}$$

The model evaluator assigns probability $1/4 + m/2$ to record value one when the model presentation is $m \in \{0,1\}$. Averaging over s_a gives the predictive law $\nu_a = \mathrm{Ber}(1/2 + a/8)$.

At each step draw independent bits ξ, η, υ with probabilities $1/4, 1/3, 1/5$ of being one, independently of previous draws, and set

$$\begin{aligned} a' &= a \oplus \xi, \\ b' &= b \oplus a \oplus \eta, \\ c' &= c \oplus b \oplus \upsilon. \end{aligned} \tag{5.6}$$

Here $\oplus$ means addition modulo two. Independently of these draws, conditional on the current state, give r' law ν_a. The record is therefore an internal state change. Its generation rule is supplied, rather than deduced from the existence of an evaluator.

First retain (a, b, r), then retain (a, r). Summing the fine transition over c' removes a normalized factor and leaves a row independent of c. Summing that row over b' removes another normalized factor and leaves a row independent of b. The configuration spaces shrink from sixteen to eight to four states. Both parents retain q_a, s_a, their evaluator, and the register. They reproduce the complete retained state and record histories for every initial law. The coupling is real, since a affects b' and b affects c'; its triangular form permits these reductions. The hierarchy itself was prescribed.

5.4 Memory and the remaining formation problem

When closure fails, a parent can retain more state or, under a declared stochastic process, the conditional law of hidden states given its accessible history. In a finite model this is a probability vector: propagate it with the fine transition, restrict to the new retained value, and normalize. That vector determines every future response even when the current summary does not. Keeping only an equilibrium hidden distribution generally changes transient predictions.

Formation requires dynamics that choose persistent blocks, hand their laws and interactions to parents, and then form another collective from at least two such parents. The next interaction must be inherited, without resetting source data or adding a newly tuned attraction. The next chapter supplies one such fusion mechanism using a nonsmooth interaction. Its additional assumptions distinguish that example from the smooth KL dynamics discussed earlier.

Chapter 6

A Nonsmooth Route to Recurrent Formation

Smooth attraction can bring two laws progressively closer without making them equal at any finite time. An absolute-difference interaction behaves differently: its force does not vanish as the gap closes. This chapter gives a constitutive extension in which four agents form two parents, and those parents subsequently fuse under the inherited interaction. The grouping follows the evolving state; the equations contain no prescribed pairs.

Pairwise convex fusion penalties and finite-time nonsmooth consensus have established precedents [2, 3]. Here the additional task is to inherit the probability laws, model evaluator, metric, and subsequent responses when fusion occurs.

6.1 Laws, a model, and inherited mass

Write $\mathrm{LN}(\mu, \sigma^2)$ for the law of a positive random variable whose logarithm is normal with mean μ and variance σ^2. A component containing n primitive agents carries

$$q_n = \mathrm{LN}(\mu^q, 1/n), \qquad s_n = \mathrm{LN}(\mu^s, 1/n), \qquad n \geq 1. \tag{6.1}$$

Recognition is the product $q_n \otimes s_n$. Its normalized reference model p_n is jointly Gaussian in the logarithms, with zero mean and precision

$$n \begin{pmatrix} a & -c \\ -c & a \end{pmatrix}, \qquad a > c > 0. \tag{6.2}$$

Thus the model couples the two sectors even though recognition is factorized. Its normalized conditional evaluator is

$$G_n(m, dk) = \mathrm{LN}\left(\frac{c}{a}\log m, \frac{1}{an}\right)(dk). \tag{6.3}$$

Direct Gaussian relative entropy gives

$$D_{\mathrm{KL}}(q_n \otimes s_n \| p_n) = C + nU, \qquad U = \frac{a}{2}\left[(\mu^q)^2 + (\mu^s)^2\right] - c\mu^q\mu^s, \tag{6.4}$$

where $C = a - 1 - \frac{1}{2}\log(a^2 - c^2)$. The mixed term makes the belief update depend on the model coordinate and conversely. The Fisher metric of this recognition family is

$$ds^2 = n\left[(d\mu^q)^2 + (d\mu^s)^2\right]. \tag{6.5}$$

Both the metric and the local potential therefore carry the same inherited mass.

A parent communicates at primitive resolution through the normalized channel

$$B_n(k, dw) = \mathrm{LN}(\log k, 1 - 1/n)(dw). \tag{6.6}$$

At $n = 1$ this means the identity channel. Adding the independent log variances shows that $q_n B_n = \mathrm{LN}(\mu^q, 1)$, with the analogous identity for s_n. This restores a primitive message law while retaining the narrower parent law internally.

6.2 The interaction that permits contact

All means here use common reference coordinates. In sector $\sigma \in \{q, s\}$, transform the source's restored message by

$$S^\sigma_{IJ}(w) = \exp\big[\chi_\sigma(\mu^\sigma_I - \mu^\sigma_J)\big]w, \qquad 0 < \chi_\sigma < 1. \tag{6.7}$$

Its log mean moves toward the receiver. Including endpoint frames U^σ_I, U^σ_J gives the displayed link $U^\sigma_I \exp[\chi_\sigma(\mu^\sigma_I - \mu^\sigma_J)](U^\sigma_J)^{-1}$, with the usual endpoint covariance under passive re-expression. The interaction depends on the laws, rather than only identifying frames.

The compared message variances are both one. Consequently,

$$\sqrt{2D_{\mathrm{KL}}\left(\widehat{\nu}^\sigma_I \| (S^\sigma_{IJ})_\# \widehat{\nu}^\sigma_J\right)} = (1 - \chi_\sigma)|\mu^\sigma_I - \mu^\sigma_J|, \tag{6.8}$$

where the displayed comparison is in a common reference presentation and $\widehat{\nu}$ denotes a restored law. Rescaling the coefficient by $1/(1-\chi_\sigma)$ gives an absolute-difference penalty with strength $\lambda > 0$. The square root is the new constitutive choice. It changes the smooth quadratic peer penalty into a convex function with a corner at equality.

On the complete contact graph, the energy apart from constants is

$$\mathcal{E} = \sum_I n_I U_I + \lambda \sum_{I<J} n_I n_J \left(|\mu^q_I - \mu^q_J| + |\mu^s_I - \mu^s_J|\right). \tag{6.9}$$

The coefficient $n_I n_J$ counts primitive contacts. With unit Fisher mobility the evolution is $n_I \dot{\mu}_I \in -\partial_{\mu_I}\mathcal{E}$. The subgradient of $|x|$ is its sign away from zero and the interval $[-1, 1]$ at zero.

Convexity makes this evolution unique. For two solutions, the derivative of their squared distance in the fixed mass metric is the negative pairing of the state difference with a subgradient difference, hence is nonpositive. Coincident primitive agents are interchangeable in the complete-graph equations. Uniqueness then keeps them equal after contact. Sticking follows from the same dynamics that brought them together.

6.3 Two successive formation events

Consider the invariant family

$$\mu_i^q = u_i + v, \qquad \mu_i^s = u_i - v, \qquad v(t) = v_0 e^{-(a+c)t}. \tag{6.10}$$

Both sectors have the same interagent gaps, so their interaction forces cancel in the equation for v. Set $\kappa = a - c > 0$ and introduce

$$u_i(t) = e^{-\kappa t} z_i(\tau), \qquad \tau = \frac{e^{\kappa t} - 1}{\kappa}. \tag{6.11}$$

This removes the common linear contraction. Before collisions,

$$\frac{dz_I}{d\tau} = -\lambda \sum_{J \neq I} n_J \operatorname{sign}(z_I - z_J). \tag{6.12}$$

Start four primitive agents at $z = (-3d, -2d, 2d, 3d)$, with $d > 0$. Their slopes are $(3\lambda, \lambda, -\lambda, -3\lambda)$. Each neighboring outer pair closes a gap d at relative speed 2λ. The resulting stages are

Time	Positions in z	Masses
0	$-3d, -2d, 2d, 3d$	$1, 1, 1, 1$
$\tau_1 = d/(2\lambda)$	$-3d/2, 3d/2$	$2, 2$
$\tau_2 = 5d/(4\lambda)$	0	4

After the first event, the parents have slopes 2λ and -2λ. Their remaining gap $3d$ closes in another $3d/(4\lambda)$. Physical event times are $t_r = \kappa^{-1} \log(1 + \kappa\tau_r)$. No coupling or mobility is retuned at the second level.

6.4 What the parent retains

At synchronization, independent log samples with a common mean have an average of variance $1/n$. Exponentiating that average gives the geometric-mean retention and the stated parent law. Restricting the primitive Fisher metric to equal location coordinates gives the factor n. Summing contact forces gives $n_I n_J$, so the reduced dynamics use exactly the inherited metric and contacts. Redundant location coordinates are removed after synchronization; an average of separated children is not claimed to close their dynamics.

Individual sample distinctions require care. If X is the retained log average, the entire primitive vector is reconstructed as

$$(X_1, \ldots, X_n) = X\mathbf{1} + R, \qquad R \sim N\left(0, I - \frac{\mathbf{1}\mathbf{1}^T}{n}\right), \tag{6.13}$$

with R independent of X. The residuals sum to zero and are correlated. Independent scalar applications of B_n to one reused parent sample preserve individual marginals but give the wrong joint law.

Fresh record sampling also closes explicitly. Applying G_n to s_n gives

$$\mathrm{LN}\left(\frac{c}{a}\mu^s, \frac{(c/a)^2 + 1/a}{n}\right), \tag{6.14}$$

the law of the geometric mean of n fresh primitive model records. A primitive record instead uses a fresh restored model sample followed by G_1. These are exact statements for the declared fresh sampling protocols; they do not establish closure for sampled-record feedback into the dynamics.

This extension establishes recurrent fusion on the common-v invariant family. It produces eventual global consensus, rather than persistent community diversity. The square-root penalty is a constitutive addition, formation under the original smooth KL dynamics remains open, and no geometry of the context base is derived.

Chapter 7

Keeping Distinct Children Under Smooth Dynamics

Exact inheritance need not identify the children with one another. The requirement is that discarded distinctions have no effect on the future quantities the parent promises to reproduce. Under smooth KL dynamics this requirement has a concrete content: hidden conditional information can contribute to the force on a retained marginal. A parent must retain that contribution, or have a reason why it cancels.

7.1 A hidden conditional law can move a visible marginal

Write a law as $q(dz, dv) = \mu(dz)\kappa_z(dv)$. Here z is retained and v is internal. Against a reference $\nu(dz)A(dv)$, the KL chain rule is

$$D_{\mathrm{KL}}(q\|\nu \otimes A) = D_{\mathrm{KL}}(\mu\|\nu) + \int H(z)\mu(dz), \qquad H(z) = D_{\mathrm{KL}}(\kappa_z\|A). \tag{7.1}$$

If one public value has a more costly hidden conditional law, changing its probability changes the full energy. The public Fisher velocity therefore contains $H(z)$ as well as the ordinary public force. A single average hidden entropy does not tell the parent which public values carry that cost.

There is an exact architecture in which this is the only hidden contribution. A sender transmits its public coordinate through a normalized, possibly law-dependent channel. The receiver reconstructs its private coordinate from its own conditional $\kappa_{i,z}$. The receiver law and compared source law then have the same private conditional, so that conditional cancels from their peer KL. With a fixed reference whose private factors are independent, the two-sector local model still couples the public belief and model laws. The original potential and Fisher metric give

$$\begin{aligned} \dot{\mu} &= -\mu[h^c + H - \mathbb{E}_\mu(h^c + H)], \\ \dot{\ell}_z(v) &= H(z) - \ell_z(v), \qquad \ell_z = \log(d\kappa_z/dA). \end{aligned} \tag{7.2}$$

The public score h^c includes every incoming interaction and every channel derivative. Cancellation is a functional identity, so it does not require freezing a live channel while differentiating.

7.2 Retain the distribution of density values

The conditional equation in eq. (7.2) moves a log-density value according to its current value and the conditional KL. It does not need the name of the private point carrying that value. This suggests

retaining the probability law

$$\xi_z = (\ell_z)_{\#}A, \qquad \int e^r \xi_z(dr) = 1, \qquad H(z) = \int r e^r \xi_z(dr). \tag{7.3}$$

For example, use a uniform reference on two private states. The laws $(3/4, 1/4)$ and $(1/4, 3/4)$ have the same spectrum: half the reference mass sees $\log(3/2)$ and half sees $\log(1/2)$. Their private preferences differ, but their contribution to the public force agrees.

The complete spectrum evolves exactly. With $a = e^{-t}$,

$$\xi_z(t) = \left(r \mapsto ar - \log \int e^{ar'} \xi_z(0, dr') \right)_{\#} \xi_z(0). \tag{7.4}$$

This follows by solving $d\kappa_z(t)/dA \propto (d\kappa_z(0)/dA)^a$ and normalizing. The formula applies to arbitrary dominated laws under the stated regularity conditions; it does not fit a Gaussian. The retained state can remain infinite-dimensional.

If all public values within an agent have the same conditional spectrum, $H(z)$ is constant and disappears under Fisher centering. The public laws then follow the original coarse KL dynamics exactly. The private conditionals can still differ: a public bit can select between the two swapped binary laws above. This gives internal correlation without changing the public force. Other agents may have different public laws and different spectra; equality between agents is unnecessary.

This property is not created at a finite time by the conditional relaxation. At every finite time $a > 0$, the power transformation and its normalization can be inverted. Different spectra therefore remain different until an asymptotic limit, unless some additional operation changes them. Retaining unequal spectra is an exact alternative, but their existence alone does not select groups of agents.

7.3 What a future record is allowed to inspect

The same two swapped private laws show why a boundary protocol matters. Let record 1 have probabilities $(3/4, 1/4)$ at the two private points. The record probabilities are respectively $5/8$ and $3/8$. Equal unmarked spectra therefore do not preserve this response.

If all permitted private readouts use a fixed mark $T(v)$, retain the joint reference distribution of $(\ell_z(v), T(v))$. A likelihood $L(o \mid z, T(v))$ then gives the conditional record probability

$$N_z(o) = \int e^r L(o \mid z, d) \eta_z(dr, dd), \qquad \eta_z = (\ell_z, T)_{\#}A. \tag{7.5}$$

After the record, the log-density coordinate changes by

$$r^+ = r + \log L(o \mid z, d) - \log N_z(o). \tag{7.6}$$

The public marginal is multiplied by $N_z(o)$ and normalized. Between records the same smooth conditional relaxation resumes. This gives an exact inherited record-and-update sequence for the declared protocols. If the marks identify every private point, however, the supposed compression may retain everything. Sufficiency is relative to what the future interaction can ask.

7.4 A smooth channel can erase a dependence in finite time

A separate four-agent example shows that exact independence can become available during the original smooth flow. Each belief and model law has a public bit, a private bit, and an internal gate bit. A fixed normalized local evaluator couples the public belief and model means. The gate means relax toward a fixed reference value. The private part of a peer channel depends on the source's private mean multiplied by a smooth function α of the population's gate means.

The function α has an exactly flat zero region. Its derivative is included in the Fisher equations. By choosing its private gain sufficiently small, the local KL force drives the gate means into this region at a finite time and keeps them there. Once inside, every peer's private cost in a receiver row is the same for all sources. It cancels from the Gibbs source comparison. The reduced energy and Fisher metric then split into a public part and an independent private-and-gate part.

The public states start in two invariant sign regions, with two agents in each. No labels are assigned to groups by the equations. Initially, private similarities can select crossed pairs, each containing one member from each public region. After private erasure, the preferred sources change: each agent prefers the other member of its public region, and those pairs persist. All cross-pair source weights remain positive. The children need not coincide; smooth uniqueness keeps distinct public states distinct at every finite subsequent time. Their separation is not claimed to stay bounded below forever.

For each pair, retain the empirical measure of its two paired public belief and model means. The original public vector field gives its exact evolution, and fresh public responses are computed by selecting and sampling a child from that measure. The private and gate laws can be forgotten after erasure. This is a noninjective reduction, but the empirical measure still retains every public child state, up to its label.

The example is an existence construction with an engineered channel, a product preparation on finite alphabets, and a specified class of boundary responses. It adds no fusion penalty and imposes no synchronization. It does not show that arbitrary smooth channels acquire a sufficient interface, that the distinct public child states admit a smaller aggregate description, or that the two resulting parents form another parent. Those are the remaining formation obligations.

Chapter 8

Links and Physical Geometry

The agent graph and the context base describe different directions of comparison. An edge $j \to i$ compares agents at one context. A path in the base $\mathcal{C}$ compares states at successive contexts. A product of communication links can be nontrivial without defining curvature on $\mathcal{C}$. Conversely, one agent can carry a curved connection over contexts; collective averaging is not required.

8.1 Relating neighboring contexts

To construct that connection, suppress the sector label and fix one agent. On a coordinate neighborhood, take a twice continuously differentiable presentation $U(c)$ and a continuously differentiable matrix-valued one-form $a = a_\mu(c)dc^\mu$. Declare directional links whose first-order behavior is

$$\Theta(c + \delta c, c) = U(c + \delta c)e^{-a_\mu(c)\delta c^\mu}U(c)^{-1} + O(|\delta c|^2). \tag{8.1}$$

The displacement enters linearly at first order, and zero displacement gives identity. These are additional properties beyond having a finite interaction matrix at each context. For a principal G-connection, the generators must belong to the represented Lie algebra; unrestricted matrices instead define general linear bundle connections.

Using $D = d + B$, expansion identifies

$$B = UaU^{-1} - dUU^{-1}. \tag{8.2}$$

The first term carries the declared directional interaction into displayed coordinates. The second compensates for the change of presentation along the path. Their relationship is exactly the operator identity

$$d + B = U(d + a)U^{-1}. \tag{8.3}$$

Indeed, acting on a vector field f gives $Ud(U^{-1}f) = df - dUU^{-1}f$; the remaining term is $UaU^{-1}f$. Squaring the operator therefore gives

$$F = dB + B \wedge B = U(da + a \wedge a)U^{-1}. \tag{8.4}$$

This is the mathematical connection to gauge theory, with the sign convention fixed by the displayed covariant derivative [8].

8.2 Curvature and paths

Curvature need not come from spatially varying coefficients. On a two-dimensional context patch choose $U = I$, $a_x = E_{12}$, and $a_y = E_{21}$. Both directional generators have rank one, but

$$F_{xy} = [E_{12}, E_{21}] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{8.5}$$

The matrix E_{ab} has one unit entry in position (a, b). Applying the two infinitesimal operations in opposite orders gives different results. This example supplies directional data; it does not derive them from the agents' laws. A presentation change alone, $B = -dUU^{-1}$, instead has zero curvature.

Finite path transport solves

$$\frac{dT}{dt} = -B(\dot{\gamma}(t))T(t), \qquad T(0) = I. \tag{8.6}$$

Its solution is a path-ordered exponential. Concatenating paths composes their transports; reversing the same path gives the inverse. Two independently specified attending edges can have unrelated generators, so their opposite labels do not impose this inverse relation. The distinction is necessary when borrowing Wilson-loop language from gauge theory [9].

8.3 Local descriptions and global compatibility

Endpoint covariance requires

$$\Theta'(c_2, c_1) = g(c_2)\Theta(c_2, c_1)g(c_1)^{-1}. \tag{8.7}$$

Differentiating the endpoints produces the affine law

$$B' = gBg^{-1} - dgg^{-1}. \tag{8.8}$$

Likewise, keeping the intrinsic transport fixed under $U' = Uh$ requires $a' = h^{-1}ah + h^{-1}dh$. A directional receiving–sending product that transforms only by conjugation is a tensorial one-form. It can supply the difference between two connections, but does not by itself supply the full affine transformation law.

Global compatibility adds another requirement. If patch coordinates obey $v_\alpha = g_{\alpha\beta}v_\beta$, then $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps, and local connection forms must satisfy the affine law there. A nontrivial bundle may have no single global presentation. Even a trivial bundle can carry flat transport with global memory: on a circle, the scalar connection $B = bd\theta$ has $F = 0$ but holonomy $e^{-2\pi b}$ around one turn. For real $b \neq 0$, this differs from identity.

Physical interpretation requires a model of the contexts, an agent mechanism producing compatible directional links, and observable responses to their transport. A connection alone does not supply spacetime or quantum dynamics.

Appendix A

Notation in Physical Terms

Symbol	Meaning and role
k, m	A represented state possibility and a model presentation. Neither symbol by itself denotes a probability law.
q_i, s_i	The agent's belief law and model law. Their marginals can evolve in coupled equations.
ζ_i	The joint law of represented state and model contents; $q_i \otimes s_i$ is a declared product restriction.
$G_i(m, dk)$	An evaluator assigning state predictions to each model presentation. Averaging gives $s_i G_i$.
$x \in \mathsf{S}$	A complete configuration of agents, including their laws and the memory and interfaces read by the dynamics.
$\rho_i : \mathsf{S}_i \to \mathsf{O}_i$	A readout of an internal distinction. Its value is a record.
Γ_{ij}	A normalized interaction transition on complete configurations. It induces record probabilities through the readout.
U_i^a	An invertible frame map for sector $a = q, s$. Changing frame changes the description.
T_j^a, O_i^a	Sending and receiving interfaces. Their product can discard directions.
L_{ij}^a, Θ_{ij}^a	A reference generator and its framed exponential interaction. The generator may be singular while its exponential is invertible.
β_{ij}, γ_{ij}	Belief-sector and model-sector source weights in normalized rows.
$\mathcal{E}$	A declared interaction potential on agent laws. Its gradient dynamics requires a metric.
$\mathcal{F}$	Variational free energy for a specified normalized probabilistic model and conditioned record.
$Q_x C$	A law of compressed represented contents. It does not specify a parent update.
$R(x), F^c$	A complete retained state and its inherited update. They must satisfy $RF = F^c R$.
$\mathcal{C}, B, F$	The context base, a connection one-form, and its curvature. These require a rule for relating neighboring contexts. The symbol F also denotes a discrete update in the inheritance chapter.

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